UI/UX External Practical Guide

Computer Laboratory II – UI/UX Design | Subject Code 417526 | SEM-I

Six practicals explained in simple steps: aim, design screens, Figma workflow, prototype interactions, expected output and viva-ready explanations.

How to use: For each practical, remember the aim, the screens you created, the main design features and how a user moves through the design. UI/UX practicals are design exercises, so there is no Arduino-style program code to memorize.

Source: Prepared from the uploaded UIUX LAB MANUAL(2).pdf, Experiments 1–6. Where the manual asks for a design output, the guide gives a simple Figma workflow rather than inventing a finished project.

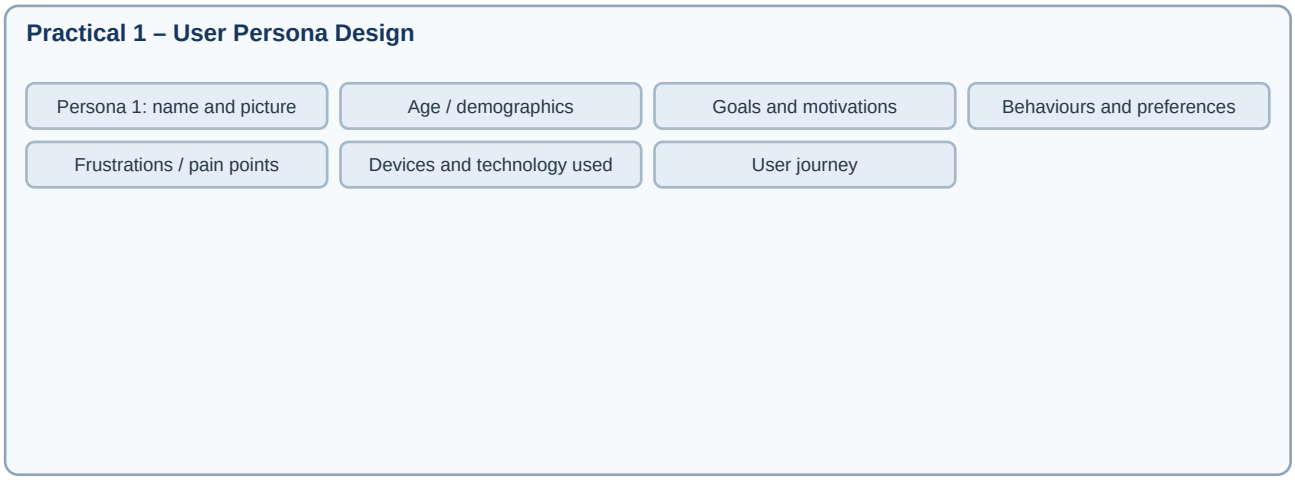
Practical 1 – User Persona Design

Aim: Design user personas for users of a selected product or system.

Simple procedure in Figma / design tool

1. User research: surveys, interviews and data about target users.
2. Identify user demographics, behaviour, preferences and technology use.
3. Create at least two personas, each with a name and picture.
4. Add demographic information, goals, motivations and frustrations.
5. Describe the user's journey while using the product.
6. Review the personas to ensure they represent different user needs.

Main screens / elements



How to explain it in the external

A user persona is a fictional, generalized representation of a real user, based on research. It helps the design team understand users and make user-centred decisions.

Expected output

At least two complete user personas with demographics, goals, frustrations and user journeys.

Likely viva questions – short answers

• Why personas? To understand target users and design for their real needs.

• Why two personas? To represent different types of users and needs.

• What is a pain point? A problem or frustration a user faces.

• What is user journey? The steps and experiences a user has while achieving a goal.

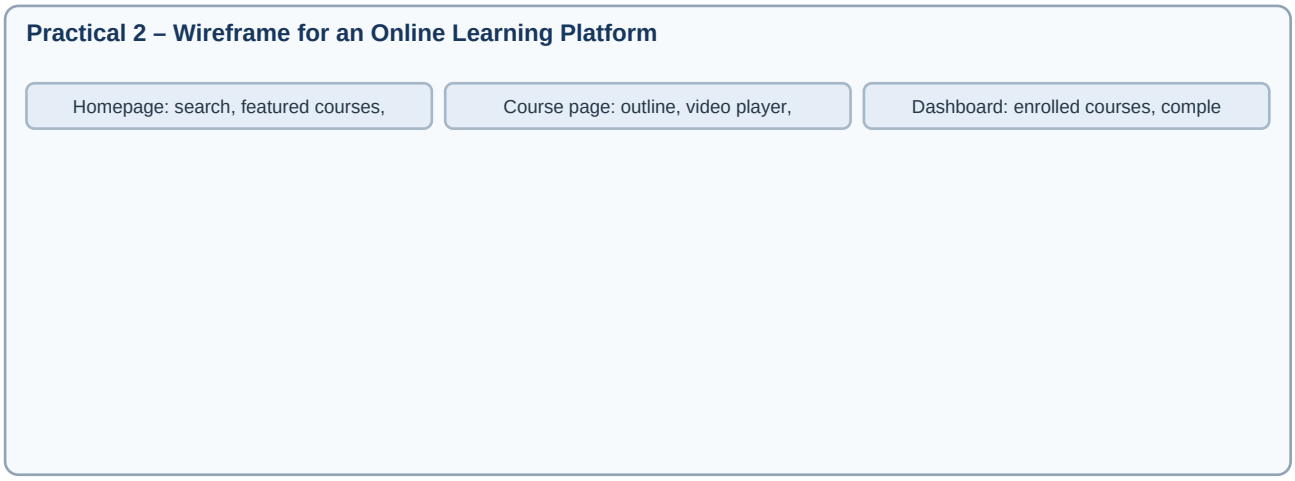
Practical 2 – Wireframe for an Online Learning Platform

Aim: Design a wireframe for an online learning platform with course listings, video lectures, quizzes and progress tracking.

Simple procedure in Figma / design tool

1. Open Figma and create a new design file.
2. Create low-fidelity frames for desktop or mobile screens.
3. Homepage: add search bar, featured courses and categories.
4. Course page: add course outline, video player, quiz access and progress.
5. Dashboard: show enrolled courses, completion percentage and milestones.
6. Connect screens in a logical user flow and review navigation.

Main screens / elements



How to explain it in the external

A wireframe is a low-fidelity blueprint of a product's layout, content structure and functionality. It focuses on structure and navigation rather than detailed colours or images.

Expected output

Wireframes of the homepage, course page and user dashboard.

Likely viva questions – short answers

• What is low-fidelity? A simple layout with limited visual detail.

• Why progress tracking? It helps learners understand their course completion.

• Why quizzes? They help learners check their understanding.

• Why wireframe first? It helps identify layout and navigation issues before detailed design.

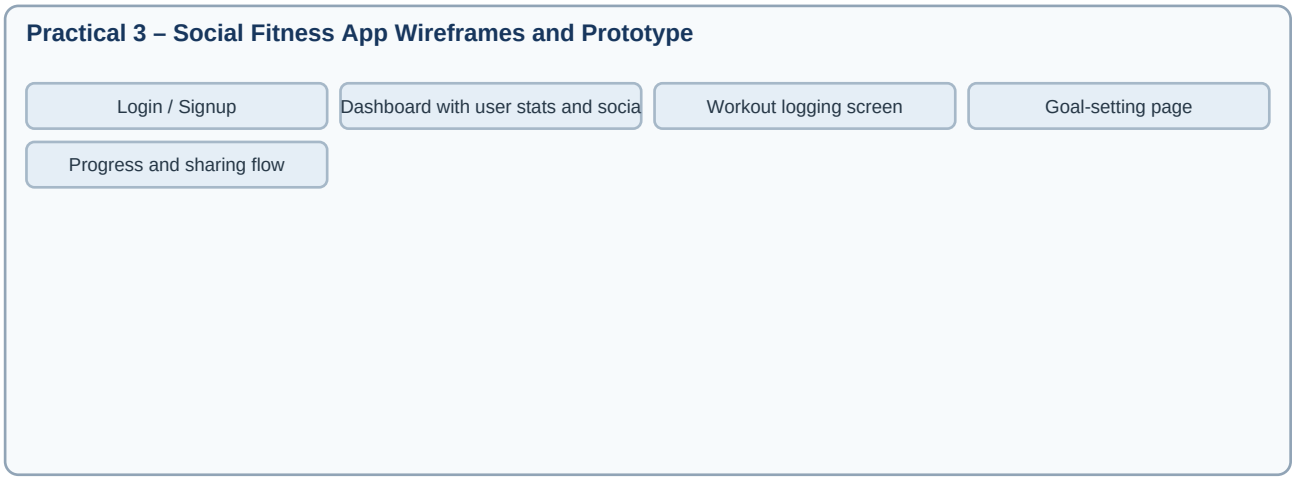
Practical 3 – Social Fitness App Wireframes and Prototype

Aim: Design wireframes and an interactive prototype for a social fitness app to track workouts, connect with friends and share progress.

Simple procedure in Figma / design tool

1. Research common fitness-app features and user needs.
2. Create login/signup and dashboard frames.
3. Add workout logging fields such as activity, duration, distance and calories.
4. Create a goal-setting page and progress display using graphs or milestones.
5. Add social feed features such as friends, likes, comments and challenges.
6. Link the frames in Figma Prototype mode to demonstrate the user journey.

Main screens / elements



How to explain it in the external

A social fitness app combines fitness tracking with social features. Wireframes show screen structure, while an interactive prototype links screens to simulate the user journey.

Expected output

Wireframes of core screens and an interactive prototype from login to workout logging and progress sharing.

Likely viva questions – short answers

• Main function? Track workouts, set goals and connect with friends.

• What metrics? Distance, calories burned and time spent.

• Why social features? They encourage motivation and accountability.

• Wireframe vs prototype? Wireframe shows structure; prototype simulates interactions and navigation.

Practical 4 – Recipe Finder Application UI

Aim: Design a user-friendly recipe finder app with ingredient search, categories, dietary filters, recipe details and favourites.

Simple procedure in Figma / design tool

1. Create a homepage with a prominent search bar.
2. Add filters for cuisine, meal type, dietary restrictions and cooking time.
3. Create recipe cards with images and clear titles.
4. Design a recipe details page with ingredients checklist, step-by-step instructions, nutrition and preparation time.
5. Add serving-size adjustment and a Save to Favorites button.
6. Create a Favorites page and simple navigation between screens.

Main screens / elements

How to explain it in the external

The recipe finder helps users discover meals using ingredients they already have and preferences such as cuisine, meal type and dietary restrictions. The UI should be simple, consistent and easy to navigate.

Expected output

A Figma design containing homepage, filters/category page, recipe details page and favorites page.

Likely viva questions – short answers

• Why search bar? It helps users quickly find recipes or ingredients.

• Why dietary filters? They narrow results to suit user needs.

• How save favourites? Use a heart icon or Save to Favorites button.

• Why clear instructions? They help users follow cooking steps without confusion.

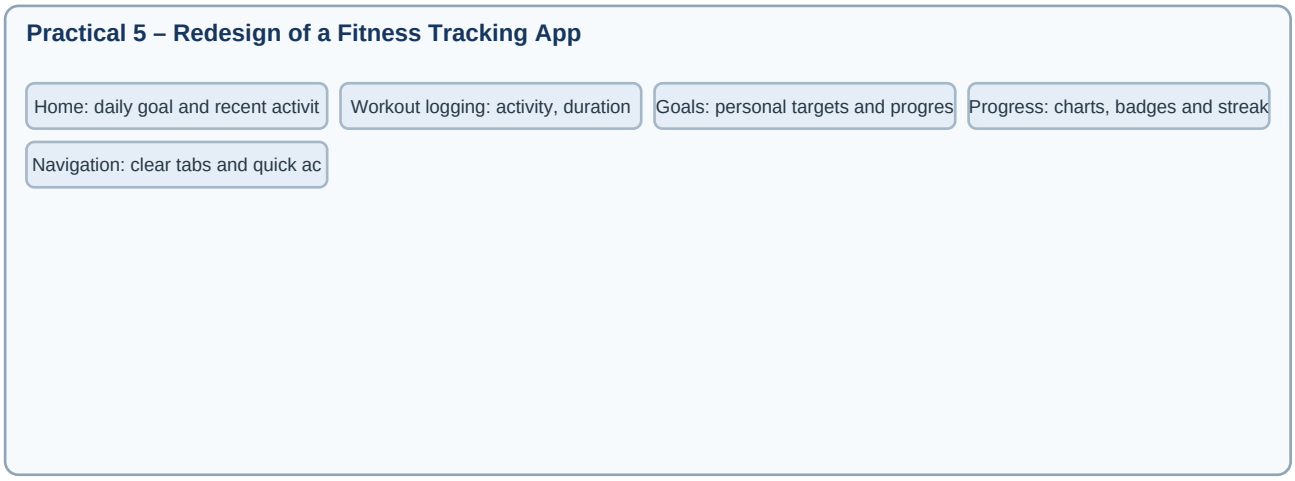
Practical 5 – Redesign of a Fitness Tracking App

Aim: Improve an existing fitness app's UI by enhancing simplicity, clarity and motivation, focusing on workouts, goals and progress.

Simple procedure in Figma / design tool

1. Review the existing fitness app and identify usability problems.
2. Redesign the home screen with daily goals and progress updates.
3. Create a quick workout logging screen with common activities and custom inputs.
4. Create a goal-setting interface with preset goals, sliders or numeric inputs.
5. Show progress through charts, completion rings, badges and streaks.
6. Simplify navigation with tabs such as Log Workout, Progress, Goals and Profile.

Main screens / elements



How to explain it in the external

The redesign focuses on making the app intuitive and motivational. It uses simple workout logging, clear goals, visual progress, reminders, badges and streaks to encourage users.

Expected output

Redesigned home, workout logging, goal-setting and progress-tracking screens.

Likely viva questions – short answers

• Why simplicity? It reduces confusion and makes tasks quicker.

• What motivates users? Badges, streaks, reminders, challenges and feedback.

• Why goal setting? It gives users clear targets to work towards.

• How show progress? Charts, graphs, progress bars, rings and achievements.

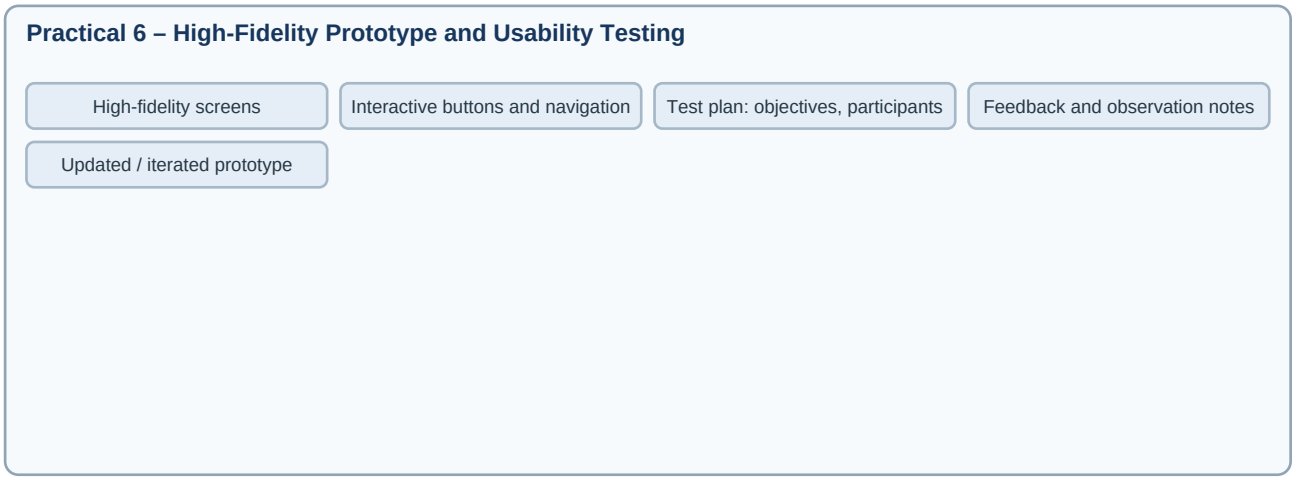
Practical 6 – High-Fidelity Prototype and Usability Testing

Aim: Develop a high-fidelity interactive prototype, conduct usability testing, collect feedback and improve the design.

Simple procedure in Figma / design tool

1. Create detailed screens in Figma using typography, colours, icons and images.
2. Connect buttons, menus, forms and navigation in Prototype mode.
3. Prepare a test plan with objectives, participants, tasks and feedback methods.
4. Ask participants to complete realistic tasks using the prototype.
5. Observe users and use think-aloud feedback; record task completion, time and satisfaction.
6. Analyze feedback, prioritize issues, refine the prototype and retest.

Main screens / elements



How to explain it in the external

A high-fidelity prototype closely resembles the final product in appearance and interaction. Usability testing checks how easily users complete tasks and identifies problems.

Expected output

An interactive high-fidelity prototype, a usability testing report and an improved prototype based on feedback.

Likely viva questions – short answers

• What is usability testing? Evaluating how easily users complete tasks.

• What is task completion rate? Percentage of tasks completed successfully.

• What is time on task? Time taken by a user to complete a task.

• What is think-aloud? Users speak their thoughts while using the prototype.

• What is iterative design? Repeatedly improving a design using feedback and retesting.

Common UI/UX Viva Quick Revision

Term	Short answer
UI	User Interface: the visual and interactive parts of a product.
UX	User Experience: the overall experience a person has while using a product.
Usability	How easily and effectively users can use a product.
User persona	A research-based fictional representation of a target user.
Wireframe	A low-fidelity blueprint showing screen structure and layout.
Prototype	A model that simulates how a product or interface works.
Low-fidelity	A simple representation focusing on structure.
High-fidelity	A detailed representation close to the final product.
User journey	Steps and experiences a user goes through to achieve a goal.
Navigation	The way users move between screens or sections.
Accessibility	Designing so people with different abilities can use the product.
Responsive design	A layout that adapts to different screen sizes.
Consistency	Using similar patterns, styles and behaviours throughout a design.
Figma	A collaborative interface-design and prototyping tool.
Usability testing	Testing a design with users to find usability problems.
Iterative design	Improving a design through repeated feedback and refinement.

External exam tip: For every practical, explain the aim, target user, screens created, key features, user flow and expected output. If asked to demonstrate, open the Figma file, select the relevant frame and use Present / Prototype mode to show interactions.